

Masruk Habib

Computer Engineer

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Profile:

Highly motivated and detail-oriented AI/ML and Research enthusiast with a strong analytical acumen and proven proficiency in Python for data manipulation and statistical analysis. Dedicated to leveraging expertise in machine learning techniques and data visualization to drive impactful insights and advance understanding in Artificial Intelligence. Eager to apply theoretical knowledge to real-world challenges and contribute meaningfully to innovative projects.

Education Qualification:

- Bachelor of Technology - Computer Engineering - 2025**
Marwadi University, Rajkot, Gujarat, India.
CGPA: 7.64 Out Of 10.00 (First Class with Distinction)
- Higher Secondary Certificate Examination - 2020 [HSC/A-Level]**
- Secondary School Certificate Examination - 2018 [SSC/O-Level]**
BIAM Model School and College, Bogura, Bangladesh.
Group: Science, GPA 5.00 Out of 5.00 (Both SSC & HSC)

Experience:

- Student Research Mentor**, Dept. of Computer Engineering, Marwadi University, Nov 2023 - Apr 2025
 - Guided 3 batches of students through the research paper writing and publication process.
 - Took 37 session in Research activity club (RAC).
 - Co-authored and published 2 research papers in collaboration with international students.
- Research Intern**, Dept. of Computer Engineering, Marwadi University, May 2024 - Jun 2024
 - Authored and published 1 research paper during the internship period.

Projects:

- MediGuard AI: An Interactive Drug and Side Effect Predictor Using LLMs with a RAG-Based Chat Assistant.** *GitHub:* <https://github.com/MediGuardAI>
 - Led the development of MediGuard AI, an LLM-driven drug interaction and side effect predictor that leverages a *fine-tuned LLM* with *data preprocessing* and *RAG-based chat assistance* to provide real-time, patient-specific medication safety insights via a *Streamlit* web application.
- UR3 CobotOps – Enhancing Cobot Efficiency with ML.** *GitHub:* https://github.com/MasrukHabib/UR3_CobotOps-Enhancing-Cobot-Efficiency-with-ML
 - Developed and implemented various ML techniques (*ARIMA, LSTM, K-Means, Classification*) to enhance UR3 cobot functionality, focusing on predictive maintenance, real-time anomaly detection, and error prediction using the UCI Machine Learning Repository dataset.
- Machine Learning Classifiers and Hyperparameter Tuning to Enhance Prediction of ASD: An Automated Django Application.** *GitHub:* https://github.com/MasrukHabib/ASD_Predictor_ML_Django_App
 - Build a Django-based predictive system for early Autism Spectrum Disorder (ASD) detection using ML algorithms (SVM, LR, ET, DT) and UCI data. Achieved perfect precision and recall via hyperparameter tuning. Published at the 2024 2nd WCONF, highlighting its innovation in healthcare AI.
- Psychological Well beingness Prediction Using ML with Hybrid Hyparameter Tuning Techniques.** *GitHub:* https://github.com/MasrukHabib/Psychological_Wellbeingness_Prediction
 - Created an intelligent ML web application for psychological wellbeing prediction, leveraging diverse classification models (Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, SVM) and hybrid hyperparameter tuning techniques (Grid Search, Random Search, Bayesian Optimization) to provide data-driven evaluations for mental health professionals.
- Fusion of Machine Learning Techniques with Hyperparameter Tuning for Hypothyroidism Prediction and Real-Time Flask Web Application.** *GitHub:* https://github.com/MasrukHabib/Hypothyroid_Predictor_ML_Flask_App
 - Collaborated on a research-based project to develop a real-time Flask web application for hypothyroidism prediction, leveraging advanced machine learning techniques and hyperparameter tuning to ensure high accuracy and reliable health assessments.
- Tableau Vizzes.** *TableauPublic:* [Profile - masruk.habib](#) | *Tableau Public*
 - Designed and deployed interactive dashboards for *Marketing Campaign Overview & Analytics, Sales Superstore History, and HR Analytics & Operational HR*, visualizing key performance indicators and driving data-driven insights.

Certifications:

- Explore Machine Learning using Python Issued by Infosys|Springboard, September 2024.
- Python Data Structures Issued by Coursera, August 2022.
- Programming for Everybody Issued by Coursera, July 2022.
- The Complete Tableau Bootcamp for Aspiring Data Scientists Issued by Infosys|Springboard, March 2023.
- Student Research Internship Training (SRIT) Issued by Marwadi University, July 2024.
- AWS Academy Graduate - AWS Academy Cloud Foundations Issued by Amazon Web Services, October 2023.
- Database Programming with SQL Issued by Oracle Academy, November 2022.
- CCNAV7: Introduction to Network Issued by Cisco Network Academy, February 2023.
- NDG Linux Essentials Issued by Cisco Network Academy, April 2023.
- Postman API Fundamentals Student Expert Issued by Postman, April 2023.
- Project Management Foundations Issued by LinkedIn, July 2023.

Skills:

Python, Machine Learning, Deep Learning, LLM, Data Analysis, SQL, Object-Oriented Programming, AWS, Software Development, Web Development, UI/UX, Interpretation Effective Communication, Microsoft Office.

Publications:

- Machine Learning Classifiers and Hyperparameter Tuning to Enhance Prediction of ASD.**
Published in: 2024 2nd World Conference on Communication Computing (WCONF).
DOI: 10.1109/WCONF61366.2024.10692172
- A Comparative Study on Hybrid Machine Learning Voting Classifier Models for Alzheimer’s Disease Prediction.**
Published in: 2023 International Conference on AI for Innovations in Healthcare.
DOI: 10.1109/ICAIIHI57871.2023.10489772
- Predicting Psychological Well-being Using Machine Learning with Hybrid Hyperparameter Tuning Techniques.**
Published in: 2024 International Conference on Artificial Intelligence and Emerging Technology (Global AI Summit)
DOI: 10.1109/GlobalAISummit62156.2024.10947775

Awards:

- Award-winning team at Azadi Ka Amrit Mahotsav Hackathon-2022.** Government of Gujarat, Mar 2023.

Awarded the winning prize at the Gujarat Azadi Ka Amrit Mahotsav Hackathon 2022, a state-level event, for an innovative project with practical impact.